

High and mighty

The world-famous Golden Gate Bridge spans a 1-mile (1.6-km) stretch of seawater, connecting the city of San Francisco with Marin County. When it opened in 1937, the bridge was the tallest and the longest suspension bridge in the world.

The vertical pillars stand 746 ft (227 m) tall and are anchored deep in the seabed.

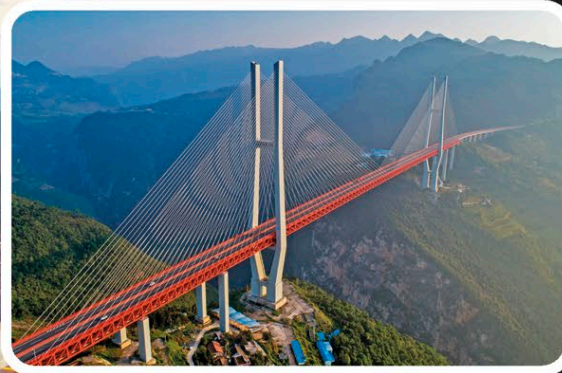
CREATING CROSSINGS

SUSPENSION BRIDGES

A bridge is a structure that supports a road, allowing people and vehicles to cross over rivers, seas, valleys, and other obstacles. There are many different types of bridges, but the design with the largest span is the suspension bridge. This is held up by long chains called suspenders, which are supported by thick cables and tall pillars. Suspension bridges are not as rigid as other types, which means they can withstand forces such as earthquakes.

CABLE-STAYED BRIDGE

Another type of bridge that uses cables to support a road is the cable-stayed bridge. Here, the cables are attached directly to the road. This picture shows the world's highest cable-stayed bridge in Guizhou, China.



TENSION AND COMPRESSION

The weight of the road and the traffic stretches the suspenders and cables—they are in tension, and they pull back, like a rubber band does, supporting the road. Where they are attached to the pillars, the cables pull downward, and the pillars are in compression. Like any object under compression, the pillars push back, supporting the whole bridge.

